

PERSISTENT SCATTERER SAR INTERFEROMETRY (PS-INSAR)
PROJECT TO ANALYSE SURFACE DEFORMATION USING
SENTINEL-1 SATELLITE DATA

INTRODUCTION

The main goal of this study is to examine and measure surface displacements in La Palma. An eruption at the Cumbre Vieja volcanic ridge, comprising the southern half of the Spanish island of La Palma in the Canary Islands, took place between 19 September and 13 December 2021. This study aims to monitor and quantify ground deformation associated with this volcanic activity using the Persistent Scatterer Interferometric Synthetic Aperture Radar (PSInSAR) approach.

Sentinel-1 satellite data are used due to their high temporal resolution and imaging capabilities under all weather conditions. The project aims to provide important insights into the temporal and spatial patterns of ground deformations caused by volcanic eruptions by utilizing Sentinel-1's capabilities. This will provide a thorough understanding of the accompanying geophysical changes.

The processing workflow integrates SNAP (the Sentinel Application Platform) for data preprocessing and interferogram generation, and StaMPS (Stanford Method for Persistent Scatterers) for advanced PS analysis and phase unwrapping. Multitemporal SAR datasets are analysed to extract reliable deformation measurements through interferometric processing, aiming for robust and validated surface displacement results in volcanic regions.

DATA ACQUISITION

The data acquisition phase involved downloading Sentinel-1 SAR images from the ASF (Alaska Satellite Facility) data portal. Sentinel-1 images are chosen for their high temporal resolution and suitability for interferometric analysis. The selection of scenes is based on the following key criteria:

- Temporal Coverage: Sentinel-1 descending pass 169 data from January 2020 to late 2023.
- Spatial Coverage: The area of interest (La Palma) is fully covered within the chosen SAR frame.

The acquired SAR data included both Single Look Complex (SLC) images for interferometric processing. Data Research using various filters which gives the data in the specific properties we need. The main filters used are File type which is Single look Complex which helps us to process interferometry as it retains both amplitude and phase information's for each pixel, the beam mode is Interferometric wide swath(IW) which provides data for extensive coverage, polarizations of the electric field in the radar wave is given as VV+VH, finally the path start and path end is the orbit number which is des 169.

MASTER SCENE SELECTION

Date	Weather	MIXR	Precipitation (Approx.)	Master Scene Selection
08-01-2022	No-event	7.85, 8.93	~0 mm	Possible
20-01-2022	No-event	3.40, 6.73	~0 mm	Possible
01-02-2022	No-event	3.28, 4.94	~0 mm	Possible
13-02-2022	No-event	8.56, 9.14	4-5 mm	Not ideal
25-02-2022	No-event	6.30, 6.61	~0 mm	Possible
09-03-2022	No-event	7.94, 9.49	2-3 mm	Possible but not ideal
21-03-2022	No-event	6.97, 7.19	~0 mm	Possible

The selection of the master scene is a critical step in SAR interferometric processing, as it serves as the reference image for generating interferograms. The master scene is chosen to ensure optimal image quality, temporal coherence, and reliable interferometric analysis. Its purpose is to minimize temporal decorrelation and maximize geometric stability across the dataset. A well-chosen master scene provides a stable baseline against which other SAR acquisitions (slave scenes) are compared, ensuring consistent deformation measurements and accurate phase unwrapping.

The master scene is chosen as the reference image and compared with the phase difference of all the other images to form an interferogram. The master scene should be selected as the image that reduces decorrelations in all interferograms. The correlation is the product four terms, which are:

- Temporal Baseline: which is time covered by interferograms, the less the time, higher the correlation.
- Perpendicular Baseline: it is the difference in satellite position at the time of scene acquisition. The less the baseline, better the results.

- Thermal Sensor noise: It is the noise that occurs due to the temperature variations. It's usually assumed as constant.
- Atmospheric influences should also be taken care. So, a month with low humidity is preferred. It can be found by analysing various Weather services and databases. Difference in doppler centroid: It is the difference in satellite position in the azimuth direction i.e., the difference in doppler centroid frequencies of the master scene and all other scenes.

For this study, 01-02-2022 was selected as the master scene based on an analysis of candidate dates. This specific date was chosen due to its favourable atmospheric conditions, including no precipitation, very low precipitable water content (5.46 mm), and stable mixing ratio (MIXR) values. These factors minimize atmospheric distortions and noise in the SAR signal. Additionally, the absence of precipitation on this date and likely preceding days reduces ground moisture, further improving the quality of the SAR imagery.

Water Vapor Analysis

Water vapor plays a significant role in SAR data quality, as it can introduce noise and distortions in the imagery. The analysis focused on several key parameters:

- Low Water Vapor Content: Scenes with minimal water vapor content are preferred to reduce atmospheric interference. Precipitable water, representing the total column of atmospheric moisture, was the primary parameter considered. Among the candidate dates, 01-02-2022 is the lowest precipitable water level.
- Consistency in Water Vapor Levels: Stability of water vapor over time is essential to maintain uniform atmospheric conditions. For the selected date, water vapor levels remained consistently low, both on the acquisition day and the days surrounding it, enhancing the reliability of the scene.

Additional Atmospheric Considerations The analysis also included precipitation data to ensure minimal moisture in the lower atmosphere and ground. No precipitation is recorded on or around the selected date, further supporting its suitability. The mixing ratio (MIXR), representing the average water vapor near the surface (entire atmospheric column), is considered as a secondary parameter. While MIXR provides localized insight into surface moisture, its relevance is secondary to precipitable water, which accounts for the total atmospheric column.

Based on the comprehensive analysis, 01-02-2022 is the most suitable master scene. It meets all critical criteria, including low and stable water vapor, absence of precipitation, and minimal geometric distortions. This date provides the optimal conditions for high-quality SAR imagery and reliable interferometric analysis.

BURST EXTRACTION

This method involves reading Sentinel-1 SAR .zip files, extracting smaller bursts from the radar images, and saving their corresponding coordinates. Each Sentinel-1 SAR image is split into bursts, which represent smaller geographic portions of the radar data and then, identifies the specific geographic location of each burst, including its corner coordinates. These extracted details are stored and then it is used for future reference and further processing steps. By breaking down the larger images, the computational load is reduced, and it becomes easier to focus on relevant portions of the image.

Defining Master Bursts

The purpose of defining master bursts is to manually select the bursts from the master image that are relevant to the area of interest. This step refines the processing scope, ensuring that the analysis focuses

on the most critical portions of the radar data. The process starts with opening the CSV file generated during the burst extraction step. Users manually delete rows corresponding to bursts that are not required for their analysis. The edited file is then saved with a new name, indicating it contains only the selected master bursts.

TOPSAR SPLIT, ORBIT FILE APPLICATION, AND CALIBRATION

The primary aim of this step is to preprocess raw Sentinel-1 SAR images to correct for satellite movement and radar errors, ensuring the data is accurate and ready for further analysis.

- **TOPSAR Split** Sentinel-1 SAR images are captured in a mode called "TOPSAR," which consists of multiple sub-swaths. The split operation divides the image into its sub-swaths to make the data more manageable for computation. Sub-swaths represent smaller areas of the radar image.
- **Orbit File Application** Satellite positions during image acquisition can deviate slightly due to orbital variations. This step applies precise orbit files, provided by ESA, to correct these positional errors.
- **Calibration** adjusts radar intensity values, converting raw data into consistent and meaningful backscatter intensity values. This ensures that radar values are accurate and comparable across different images and datasets, providing physically meaningful measurements for further analysis.

How the Script Works

1. **Data Handling** The script processes all Sentinel-1 SAR .zip files located in the specified data directory. Each file is handled iteratively to ensure consistent processing.
2. **Modify SNAP Graph** A predefined graph file (graph_split_orb_cal_snappy.xml) contains step-by-step instructions for processing the data. The script modifies this graph file to include the correct input/output file paths and parameters, such as sub-swath and burst details.
3. **Process Execution** The gpt (Graph Processing Tool) command-line utility is used to execute the updated graph file for each image. The tool carries out the split, orbit correction, and calibration steps automatically.

The outputs of this process are .dim files, such as YYYYMMDD_split_orb_cal.dim. These files represent radar images that have been split into sub-swaths, corrected for orbit errors, and calibrated, making them ready for advanced geospatial and interferometric analysis.

COREGISTRATION

In Synthetic Aperture Radar (SAR) imaging, satellites orbit the Earth and capture radar images of the same area over time. However, slight variations in satellite position or orientation mean that the same ground location may appear in different pixel positions across images. Coregistration is a crucial preprocessing step that addresses these variations to ensure accurate analysis.

Coregistration Achieves:

- **Spatial Alignment** aligns the pixels of the slave image to match the pixels of the master image. This is essential for deformation measurements, which rely on precise comparisons of the same location across images.
- **Spectral Alignment** corrects for minor differences in radar frequency or signal phase between images. Without this correction, measurements might be inaccurate.

The process involves two primary steps: Back-Geocoding and Enhanced Spectral Diversity (ESD).

- **Back-Geocoding** aligns the geographic positions of the slave image to match the master image. It utilizes satellite orbit data and a Digital Elevation Model (DEM) to adjust each pixel in the slave image to its corresponding pixel in the master image.
- **Enhanced Spectral Diversity (ESD)** After Back-Geocoding, slight spectral misalignments may remain due to differences in radar frequency or signal phase. ESD fine-tunes the alignment using the radar signal's properties.

How the Script Works

1. **Data Handling** The script scans the data directory (data_dir) for Sentinel-1 SAR .dim files. It identifies the master image based on the specified master date and pairs it with the corresponding slave images.
2. **Modify SNAP Graph** A predefined graph file (graph_coregistration.xml) contains instructions for coregistration. The script dynamically updates this graph file to include the correct file paths for the master and slave images. The output file paths are also customized for each processing iteration.
3. **Process Execution** The script uses the Graph Processing Tool (gpt) from the SNAP toolbox to run the updated graph file. Each slave image is processed against the master image, ensuring accurate spatial and spectral alignment through Back-Geocoding and Enhanced Spectral Diversity (ESD). Log files capture the progress and any errors during execution.

Coregistration is Important for InSAR Interferometric Synthetic Aperture Radar (InSAR) relies on comparing the phase information of SAR images taken at different times. Coregistration is critical because, if pixels representing the same ground point are not aligned, the phase differences will be incorrect, resulting in unreliable deformation measurements. Coregistered images ensure that phase differences are due to ground movement, not misalignment or noise, allowing for precise and meaningful analysis of ground deformation.

INTERFEROGRAM FORMATION AND STAMPS EXPORT

This step involves generating interferograms from coregistered SAR images and preparing the data for time-series deformation analysis using StaMPS. It is a critical stage where the phase differences between the master and slave images are calculated to detect ground deformation over time. An interferogram is a visual representation of the phase difference between two SAR images (master and slave). This phase difference can provide information about ground deformation, topographic variations, and atmospheric effects. The goal in this step is to isolate ground deformation by removing the topographic and atmospheric contributions.

The process begins by forming interferograms, which involves combining the master and each slave image to compute the phase differences. The topographic phase is removed using a digital elevation model (DEM) to eliminate terrain-related contributions. The resulting interferograms are then cropped to a specific region of interest, and the data is exported into a StaMPS-compatible format for further time-series analysis.

How the Script Works

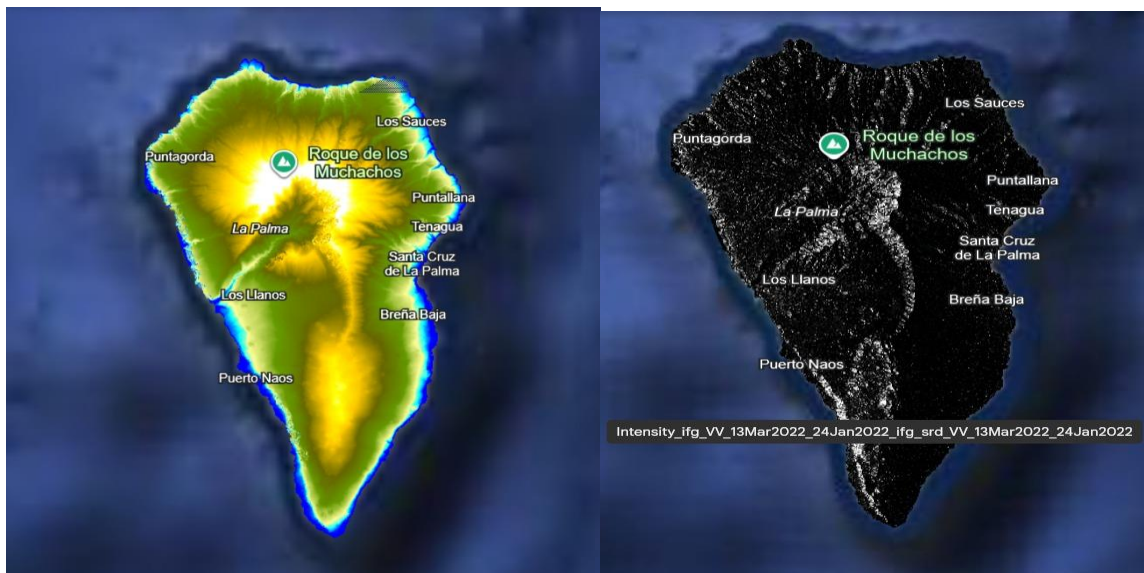
1. **Reading Input Data** The script scans the Coregistration directory for all .dim files, which represent the coregistered SAR images. It extracts the dates from the filenames to create a list of master-slave pairs for processing.

2. **Preparing Output Directories** The script ensures the necessary directories for storing interferograms (Intfs) and StaMPS exports (Exports) are created and organized.
3. **Modifying the SNAP Graph** The interferogram formation, topographic phase removal, and StaMPS export steps are defined in a SNAP XML file (graph_ifg_export.xml). This XML file is dynamically updated by the script to include, Input file paths (e.g., 20220201_split_orb_cal_backgeo_esd.dim).Output paths for the interferograms and export folders. Subset coordinates (geographic or pixel values) to focus on the region of interest.
4. **Running the Interferogram Process** The SNAP gpt tool is used to execute the updated graph file. It generates interferograms from the master-slave pairs. Applies subsetting to limit the interferograms to the area of interest. Exports the interferograms and associated metadata into a StaMPS-compatible format. The script logs the progress of the process, including error messages and completion notifications.

Interferograms highlight ground deformation by displaying phase differences between image pairs. By removing the terrain-related phase, the data focuses solely on ground deformation. The exported files are ready for advanced time-series deformation analysis, enabling the detection and monitoring of ground changes like subsidence or uplift.

OUTPUTBANDS

This step involves analysing the interferogram outputs generated in the previous step. Users open the results in the Interferograms directory and export key bands such as intensity, phase, coherence, and elevation as KMZ files for visualization in Google Earth. The task is to identify correlations between coherence, intensity, elevation, and land usage, such as urban areas, forests, or water bodies. Users also examine the phase images for recognizable structures and evaluate whether the study area is satisfactorily covered. This visual inspection ensures that the outputs are meaningful and ready for time-series deformation analysis.



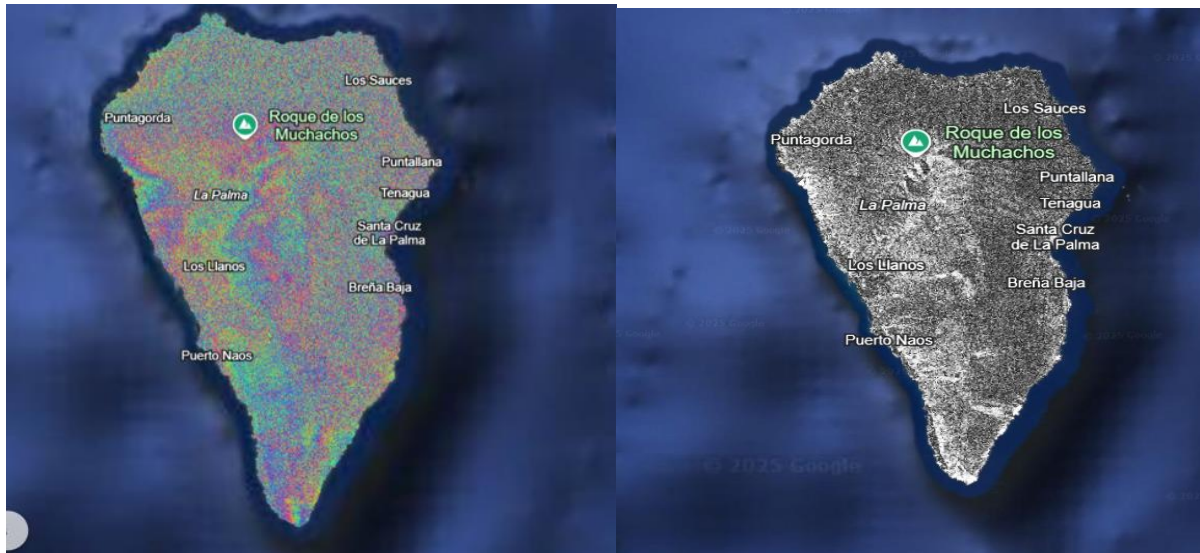


Fig. Elevation, Intensity, Phase image and coherence images as output bands shown Google Earth Pro.

The elevation map (Figure A) illustrates La Palma's diverse topography, with the highest elevation clearly marked at Roque de los Muchachos in white and yellow tones. Lower regions such as coastal zones appear in green to blue shades, reflecting their lower altitudes. This topographic data is particularly important for interferometric processing, as it helps correct phase distortions induced by elevation differences, ensuring more accurate deformation results.

The radar intensity image (Figure B) captures the strength of backscattered radar signals. Brighter areas, such as those seen in Puerto Naos, Breña Baja, and Santa Cruz de La Palma, correspond to urban regions or rocky terrain where surfaces reflect radar waves more efficiently. In contrast, darker areas generally indicate vegetation or water bodies, which absorb or scatter radar energy. These high-intensity zones are essential for identifying Persistent Scatterers (PS), as they maintain consistent radar reflectivity over time.

The wrapped interferometric phase map (Figure C) displays the phase difference between two radar acquisitions. The colourful fringes indicate phase shifts that may result from topographic changes, surface displacement, or atmospheric variations. While these fringe patterns are valuable for detecting potential deformation zones, they must be unwrapped and corrected to extract meaningful displacement data. The map provides an early visual cue for detecting ground movement, especially in areas affected by volcanic activity.

Lastly, the coherence map (Figure D) measures the consistency of radar signals between acquisitions. High coherence is shown in bright regions, typically urban or rocky areas with minimal temporal change, while low coherence appears in darker areas such as forests or agricultural fields. This map is crucial for PS selection, as only pixels with stable coherence are suitable for long-term deformation monitoring. High coherence areas serve as reliable anchors in subsequent time-series analysis using StaMPS.

When analysed together, these four output bands enable a comprehensive understanding of surface characteristics, stability, and the potential for deformation. They support accurate selection of PS points, allow corrections for topography and atmospheric effects, and guide interpretation of surface displacement across the island.

SNAPEXPORT

During this step, users explore the Exports directory created by script 06. The task includes identifying subdirectories and file types, such as .par files, and comparing their numbers to those in the Interferograms directory. By examining a .par file, users extract parameters like heading, range pixel spacing, and azimuth pixel spacing, interpreting their significance for interferometric analysis. These parameters are then cross-checked against metadata in SNAP for consistency. This step ensures that the exported data is accurate and well-prepared for StaMPS processing.

PREPARATION FOR STAMPS PROCESSING

The final step involves preparing the interferometric data for StaMPS analysis. Users create a dedicated folder for StaMPS processing and run the `mt_prep_snap` command, customizing the master date and data path parameters. This step marks the transition from SNAP-based processing to StaMPS, setting up the framework for detecting persistent scatterers and conducting time-series deformation analysis. It is the culmination of all previous steps, ensuring that the data is in the required format and structure for advanced geodetic applications.

OUTLIER DETECTION AND DATA ANALYSIS

After preparing the data using the `mt_prep_snap` command, the results are evaluated to ensure data quality before continuing to StaMPS processing. This included the detection of outliers, statistical evaluation of the mean amplitude, and the visualization of interferograms.

Mean Amplitude Analysis

The `calamp.out` file, generated during the preparation step, contains the mean amplitude values for each scene. Mean amplitude analysis is critical for identifying scenes with abnormally low values, which might indicate corrupted or poor-quality data.

To systematically detect anomalies, the MATLAB script `outliers_amplitude.m` is executed. The following outputs are generated:

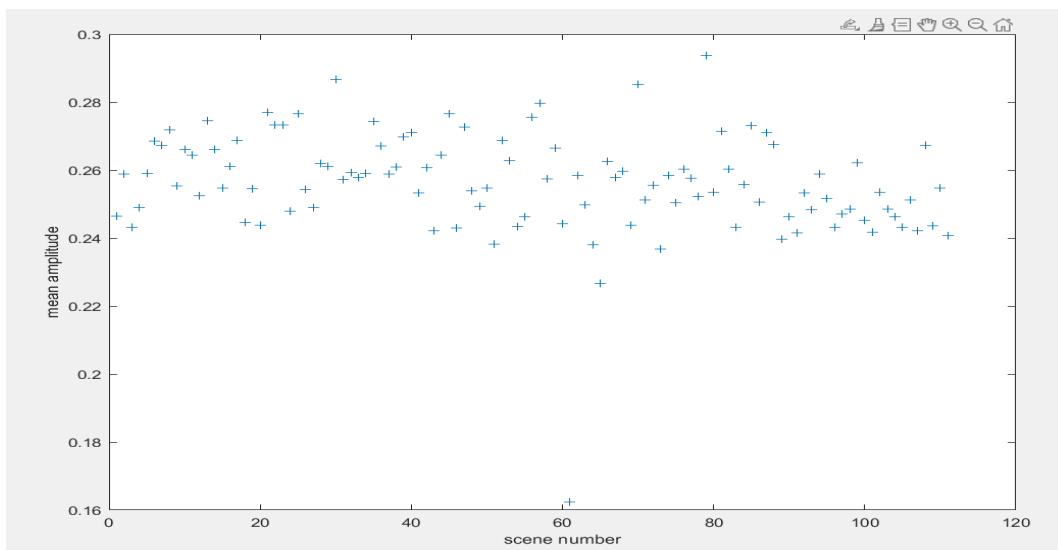


Fig 1. Mean Amplitude vs. Scene Number

The scatter plot shows the mean amplitude values for all scenes, with the x-axis representing the scene number and the y-axis representing the mean amplitude. Most scenes fall within a consistent range of

mean amplitude (0.24–0.28). Although Scene 61 was identified as a statistical outlier due to its lower mean amplitude (~ 0.16), it was not removed from the dataset. This is because Scene 61 serves as the master scene, and the reduced amplitude is attributed to a processing artefact—specifically, ocean pixels being set to zero in the master scene but not in others. As a result, the lower amplitude does not reflect a real quality issue in the scene itself. Based on this context, the scene was retained in further analysis despite its outlier classification.

Boxplot of Mean Amplitude

The boxplot provides a statistical summary of the mean amplitude values. The central box represents the interquartile range (IQR), with the red line indicating the median. Outliers are plotted as individual points. Scene 61 appears as an outlier below the lower whisker, confirming its anomalous amplitude value.

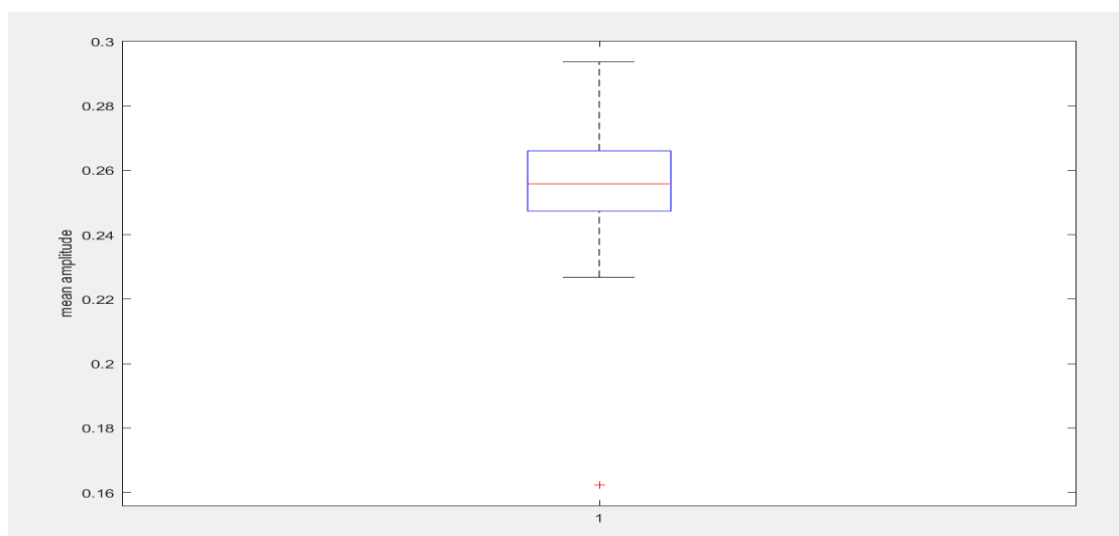


Fig 2. Boxplot of Mean Amplitude

Histogram of Mean Amplitude Distribution

A histogram shows the frequency distribution of the mean amplitude values. Most scenes exhibit mean amplitudes between **0.24–0.28**, while a few scenes fall below this range, notably scene 61. The log-scaled histogram highlights the overall consistency of the data, with minimal extreme values.

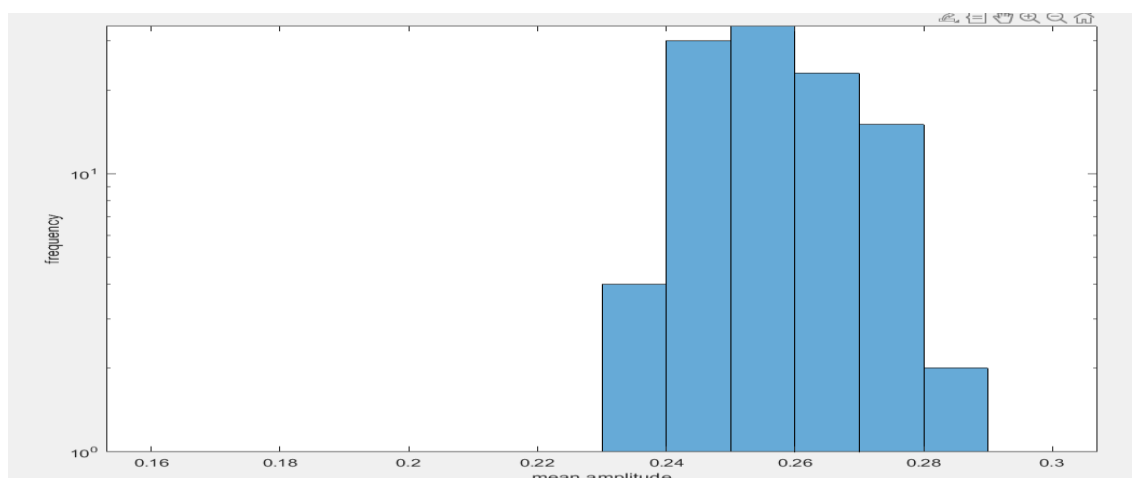


Fig 3. Histogram of Mean Amplitude Distribution

STAMPS PROCESSING

The PS Candidates are chosen according to the Amplitude Dispersion Index D_A , which is a measure used to select pixels that are likely to be stable over time.

$$D_A = \frac{\sigma_A}{\mu_A}$$

σ_A and μ_A are temporal standard deviation and the temporal mean.

The reason why we use amplitude and not phase is due to the phase ambiguity, which means that the phase is still wrapped in cycles of 2π .

$$D_A \approx \sigma_\phi$$

σ_ϕ = Standard deviation of phase

$D_A = 0.42$

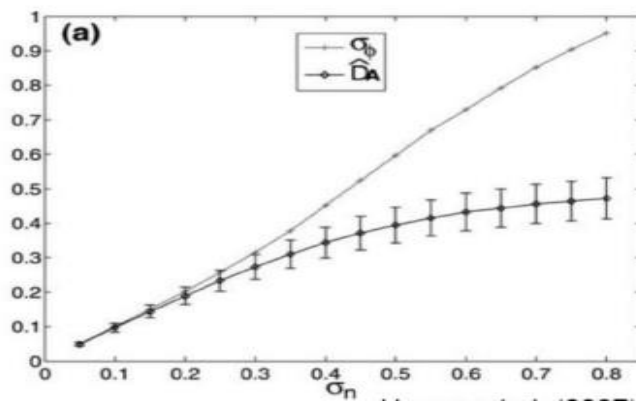


Fig 4. The mean values of D_A are plotted for each value of σ_n , as in Ferretti et al. [2001], with error bars representing one standard deviation. Also plotted are corresponding values of phase standard deviation, σ_ϕ . [Hooper et al. 2007]

The StaMPS (Stanford Method for Persistent Scatterers) processing is divided into multiple steps to identify and analyse Persistent Scatterers (PS) in Synthetic Aperture Radar (SAR) data. This section explains **Steps 1 to 5** and highlights the outputs required for the report.

Initial Data Preparation and Baseline Estimation (Step 1)

The PS-InSAR processing began with Step 1, which involved preparing the input data and estimating the spatial and temporal baselines for the interferograms. Sentinel-1 SAR images over La Palma are pre-processed to generate interferograms, with careful attention paid to co-registration accuracy. The temporal baseline ensured that interferograms are generated over a consistent observation period, while the spatial baseline optimized the geometric alignment between SAR acquisitions. This step also involved configuring the necessary parameters, such as `ifg_proc` and `master_scene`, to define the master and slave images used for interferogram generation.

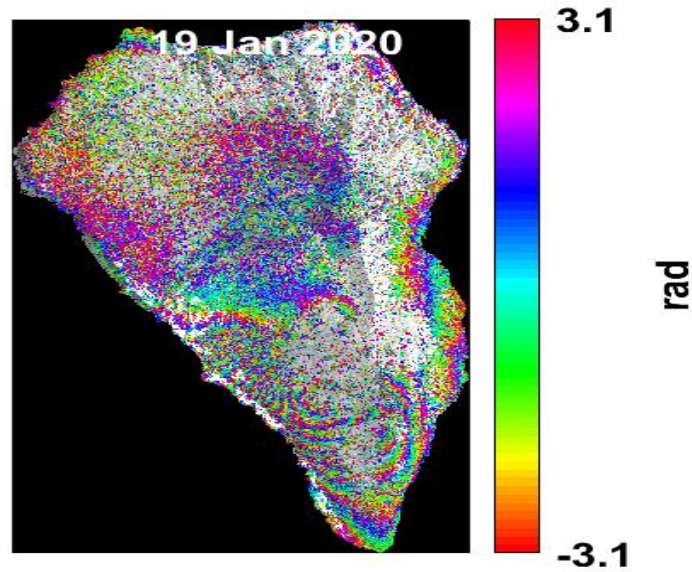


Fig 5. Wrapped interferogram

Phase Noise Estimation (Step 2)

Phase Noise Estimation (Step 2) Step 2 of the StaMPS processing focused on estimating phase noise in the interferograms to identify potential Persistent Scatterer (PS) candidates. The phase signal is decomposed into components such as deformation, topography, atmospheric delays, and noise to isolate pixels exhibiting stable phase characteristics over time. Low-noise pixels are flagged as preliminary PS candidates and stored for further analysis. To optimize this step, parameters controlling noise thresholds were adjusted iteratively. Challenges included distinguishing between phase variations due to deformation and those caused by noise, which are mitigated through careful threshold selection. The results are logged in the STAMPS.log file, providing a summary of the PS candidate selection process.

Persistent Scatterer Selection (Step 3)

PS Selection (Step 3) In Step 3, the selection of Persistent Scatterers is refined by applying coherence thresholds and evaluating phase stability metrics. The stability of each pixel's phase is compared to a reference noise model, and only the most stable pixels were retained as PS points. Key parameters such as density_rand are adjusted to ensure a uniform spatial distribution of PS points while maintaining high coherence. Outputs of this step included the PS Candidate List, detailing the spatial distribution of selected Persistent Scatterers. The ps_plot('w') command is used to visualize the distribution and confirm the selection quality. Challenges in this step involved balancing the density of PS points with their stability, which is resolved through iterative threshold adjustments and visual verification of results.

Phase Unwrapping Preparation (Step 4)

Phase Unwrapping Preparation (Step 4) Step 4 prepared the selected PS candidates for phase unwrapping, a critical step in converting wrapped phase values (modulo 2π) into continuous phase measurements. During this step, problematic PS points with high phase discontinuities are identified and excluded. Outputs included refined PS maps with updated phase values and unwrapping preparation logs detailing the adjustments made to key parameters. Challenges encountered included handling phase inconsistencies in areas with low coherence, which are addressed through targeted parameter tuning and iterative refinement of the PS candidate list.

Weeding Parameters and Analysis

The refinement of weeding parameters played a pivotal role in improving the quality of the PS selection. A histogram of the standard deviation of PS points is generated to analyse the distribution of phase noise, providing insights into optimal threshold values for weeding. The `weed_standard_dev` parameter is adjusted iteratively to exclude high-noise points while retaining stable scatterers.

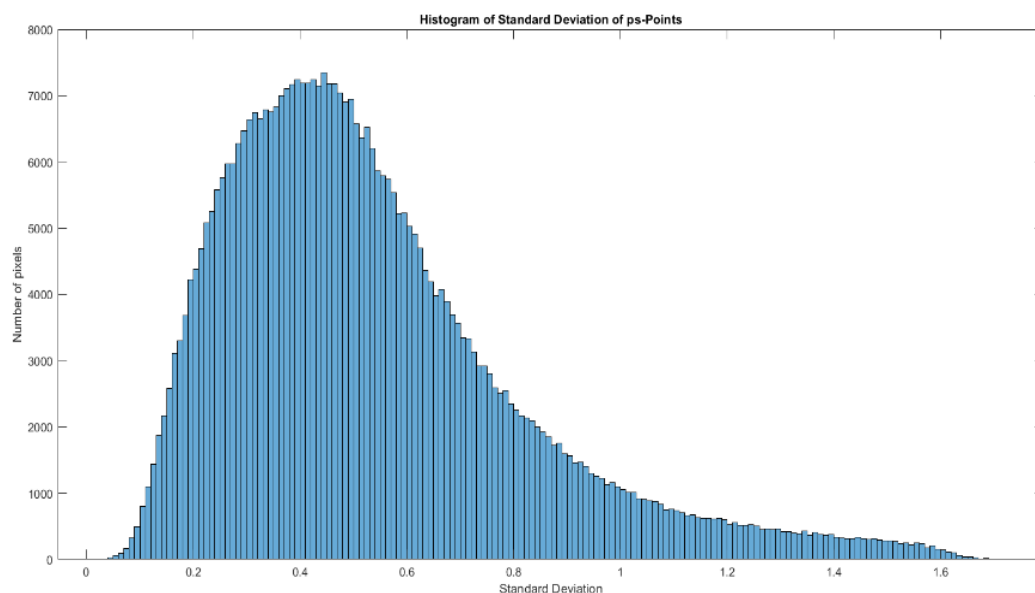
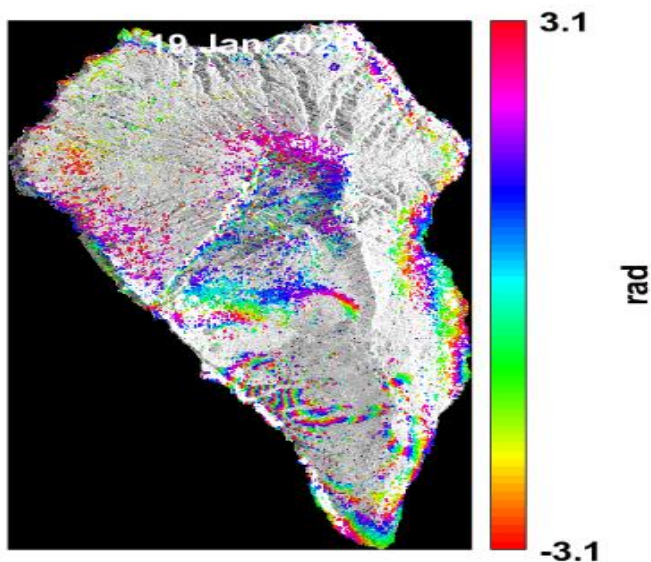


Fig 6. Histogram of standard deviation of ps points



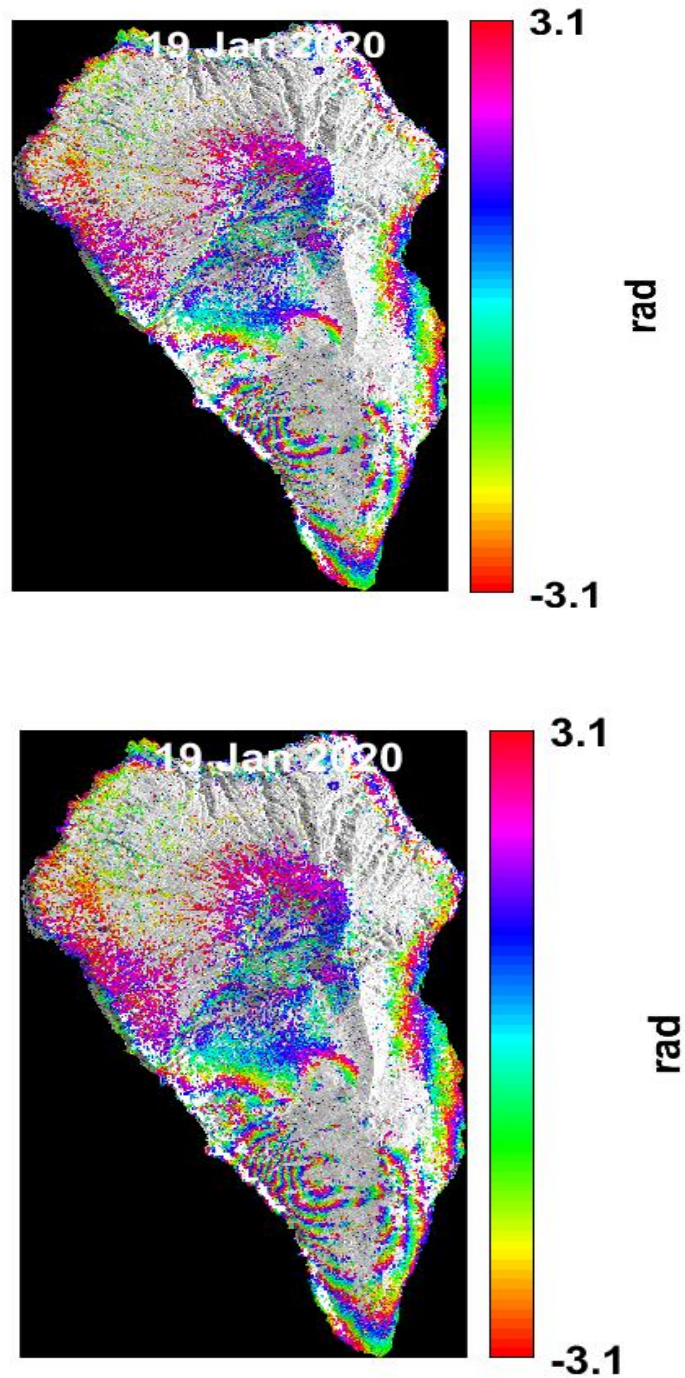


Fig 7. Wrapped interferogram of different standard deviations 0.4,06,0.8 respectively

Initial Phase Unwrapping and Refinement (Steps 6-8)

Phase unwrapping was conducted in Steps 6 through 8 to derive continuous phase values from the interferograms. The "3D" unwrap method is applied with a grid size of 100 meters, and parameters such as `stc_w_thresh` were fine-tuned to enhance the results. Initial unwrapping results revealed areas with significant noise and phase discontinuities, which are further analysed using `investigate_uw.m`. Problematic interferograms and scenes, such as indices 249, 250, and 251 etc. are identified and removed

using the `drop_ifg_index` and `delete_ifg` commands. Weeding thresholds, including `weed_max_ncp_thresh` (set to 18) and `weed_av_sum_ncp_thresh` (set to 2.5), were adjusted to improve the quality of the unwrapped phase maps. These refinements are followed by another iteration of stamps (6,6), producing improved results.

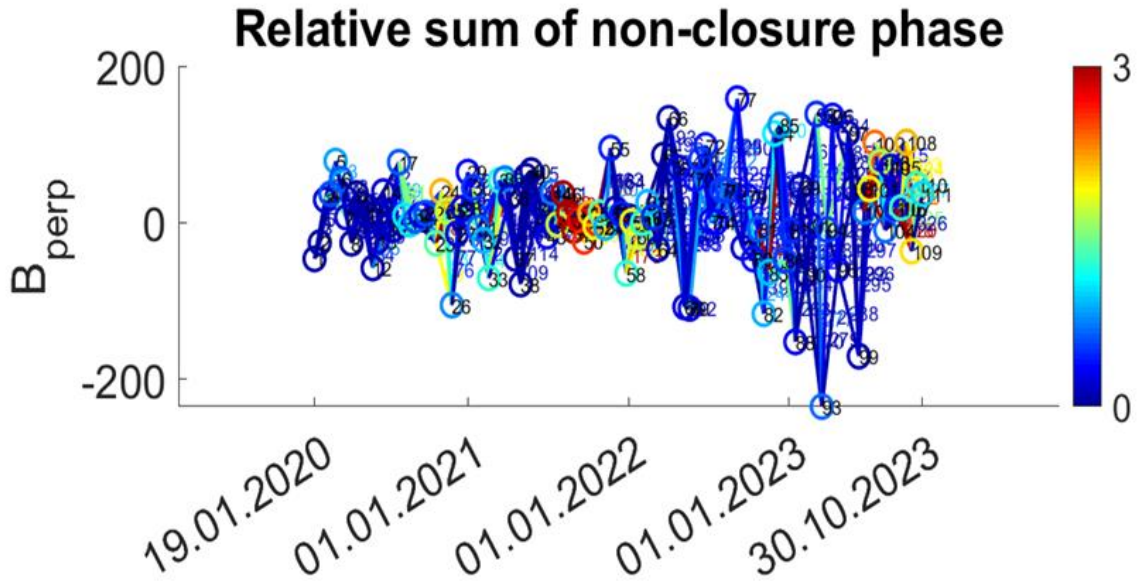


Fig 8. Relative sum of non-closure phase

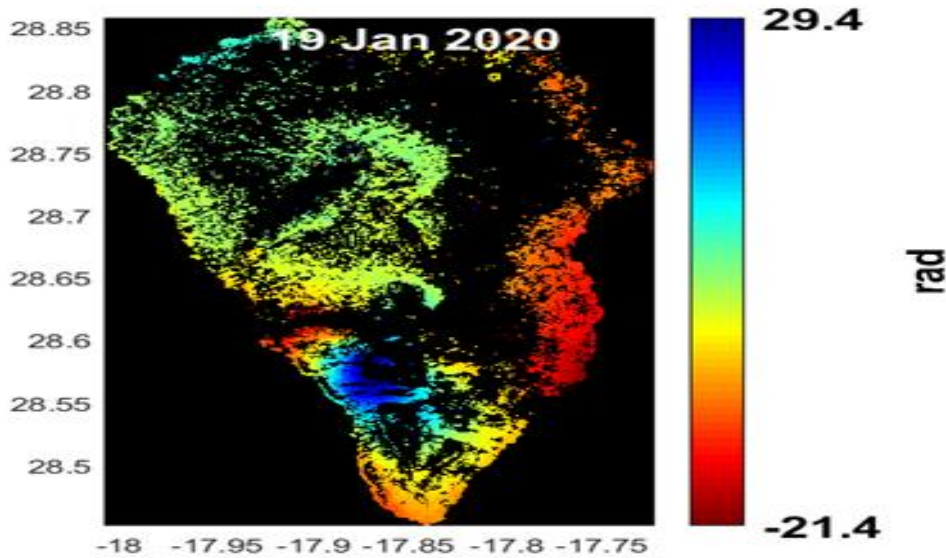


Fig 9. Unwrapped interferogram

The unwrapped phase results are further investigated using the `investigate_uw.m` script. The script allowed for the visualization of problematic areas, enabling the identification of any remaining inconsistencies. Figure 5, 6 were included to provide a comprehensive overview of the results. The relative sum of non-closure phases is plotted to identify interferograms with residual inconsistencies, which are subsequently addressed by refining the processing parameters.

Iterative Refinement and Velocity Analysis

Stamps_iter was used to analyse results from steps 6 to 8. Input parameters were adjusted, including ref_centre, velo_rg, and idx_ifg, to tailor the visualization to areas of interest and ensure accurate color scaling. The reference area selected for La Palma was ref_centre = [-17.9125, 28.66], located in a region with minimal displacement as verified via GNSS and EGMS data.

Phase Analysis and Correction (Steps 14–15)

After phase unwrapping, Steps 14 and 15 focused on analyzing and refining the phase components to isolate the deformation signal. The **ps_plot** tool was used with multiple abbreviations to separate different error sources contributing to the wrapped phase: The phase analysis and correction began by visualizing the **unwrapped phase** using the command ps_plot('u',...)

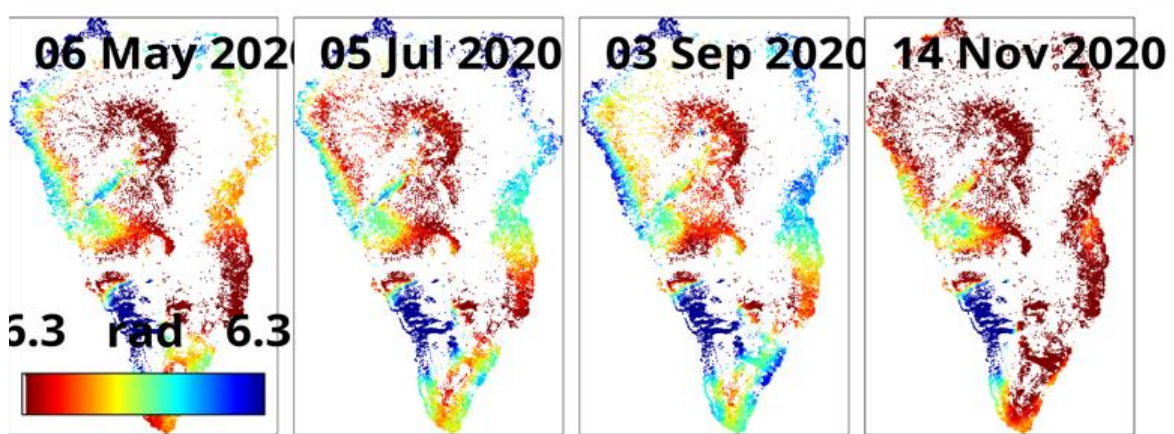


Fig10. Unwrapped phase visualization

The **mean velocity** of Persistent Scatterers (PS) was plotted using ps_plot('v',...), providing insight into the line-of-sight (LOS) deformation rates across the study area in mm/year.

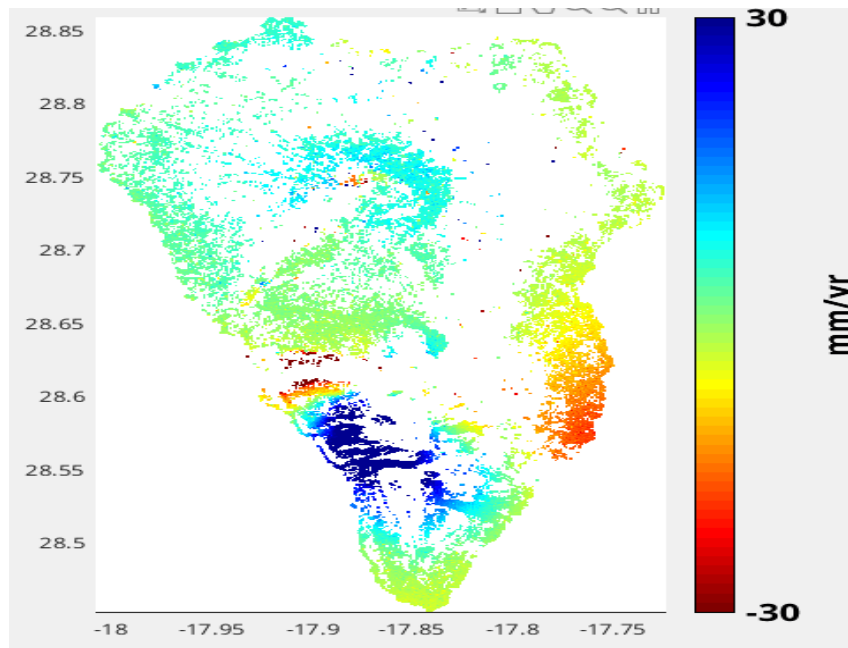


Fig11. Initial mean velocity (contains atmospheric/orbital noise)

This step was crucial for identifying zones of uplift or subsidence, although at this stage, the velocity data still contained errors due to atmospheric and orbital influences. To further decompose these errors, specific components were analyzed using `ps_plot('s',...)` and `ps_plot('o',...)`. These commands respectively visualized the slave atmosphere and slave orbital ramps, helping to assess the spatial extent and intensity of these error terms.

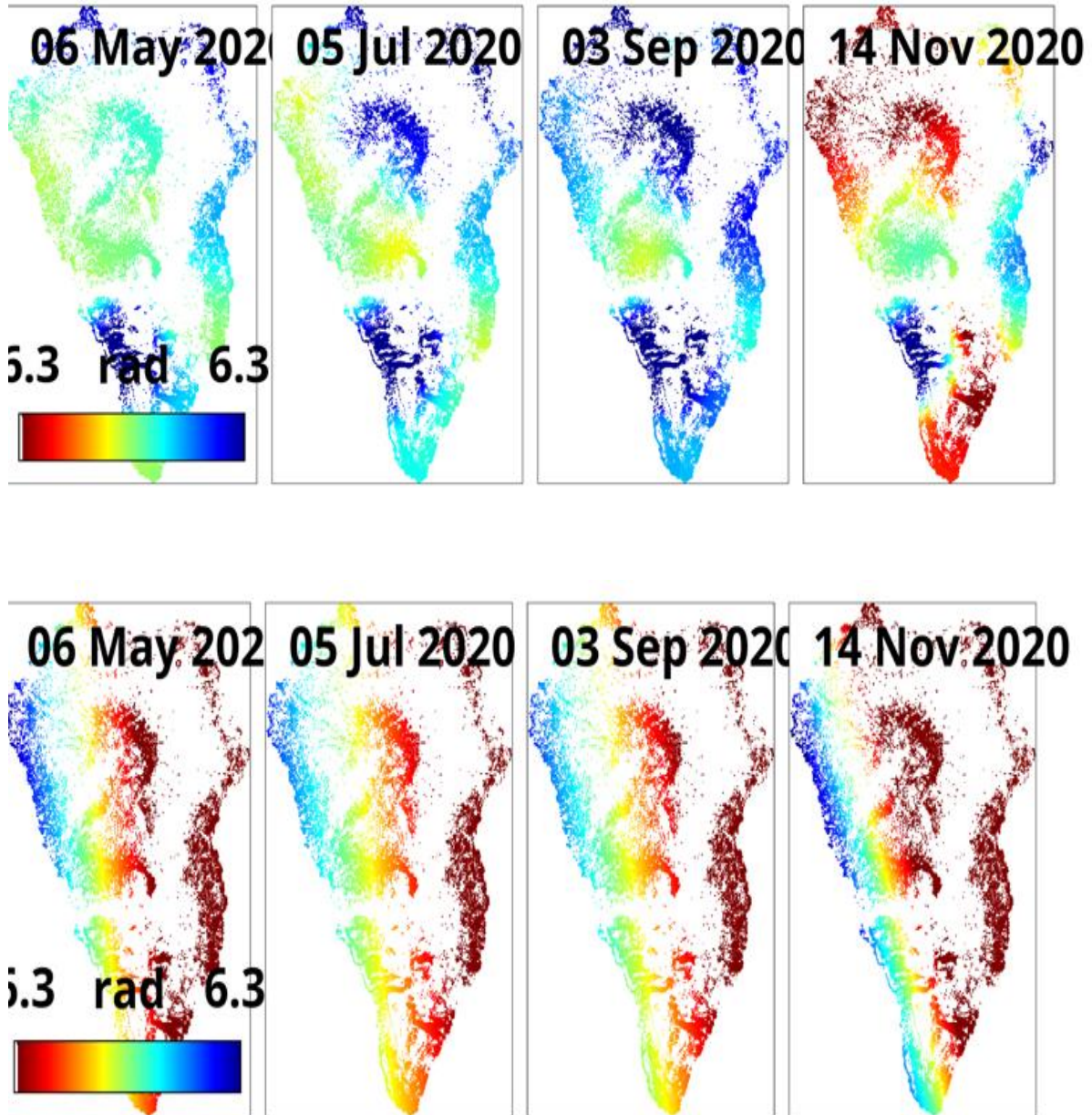


Fig12,13. Slave Atmosphere and Orbital Ramp Error Maps

Finally, the corrected mean velocity was visualized using `ps_plot('v-dso',...)`, which displays the velocity after the removal of atmospheric delays, orbital ramps, and look angle errors. This refined visualization isolated the true ground deformation, providing a cleaner and more reliable deformation map for interpretation and further geophysical analysis.

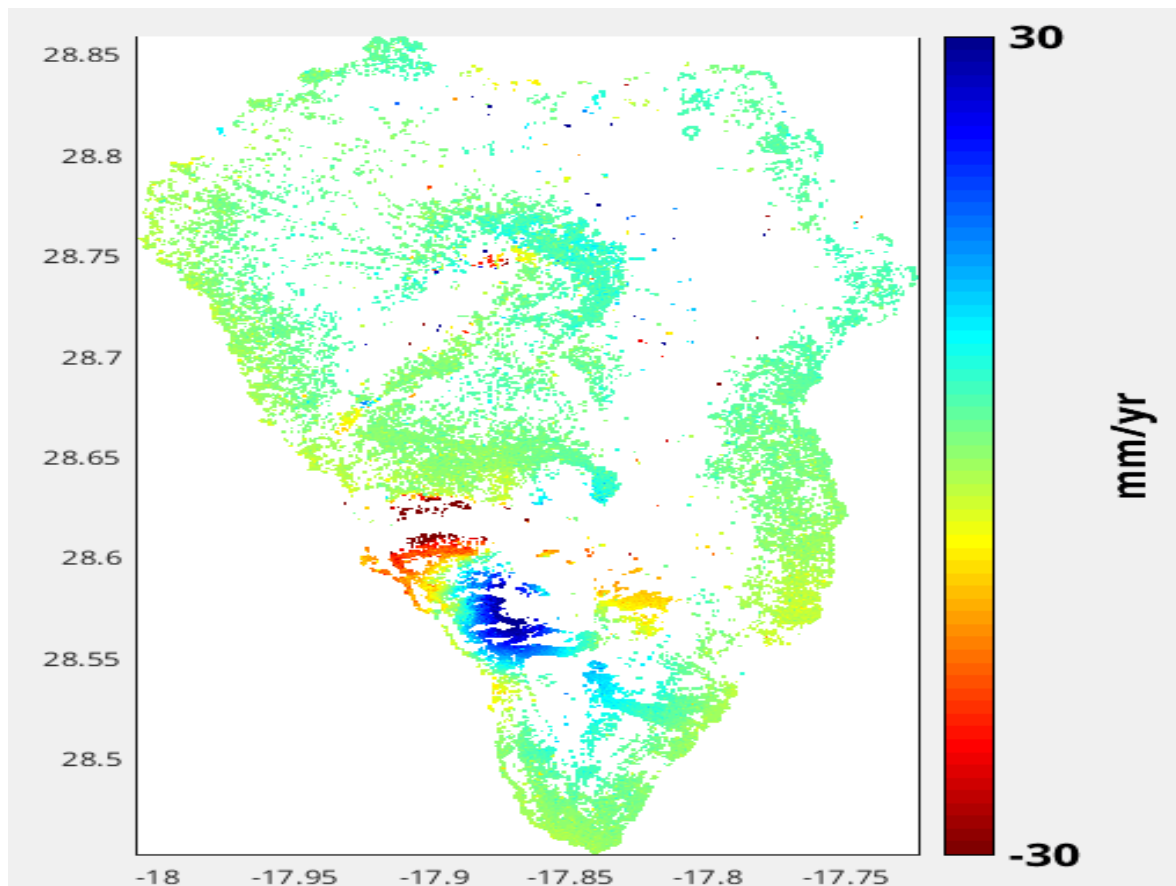


Fig 14 Progressive Mean Velocity Correction Maps

Key Observations:

- Post-Correction Clarity:**
 The v-dso maps reveal a concentrated deformation zone in the southwestern part of the island, near the Cumbre Vieja ridge. This area shows uplift velocities approaching +30 mm/year, likely caused by magma intrusions and volcanic activity during the 2021 eruption.
- Validation:**
 Non-deforming zones, such as the northern and eastern coasts, show little to no movement post-correction, confirming the accuracy of atmospheric and orbital error removal. Visual inspection supports that displacement is localized around volcanic areas.
- Overall Impact:**
 The combination of phase corrections and the revised colour scale enables clear identification of actual ground deformation. This improved interpretation is critical for hazard assessment and validates the effectiveness of the PS-InSAR technique.

Mean Velocity Analysis (Step 16)

In Step 16, mean velocity maps were generated to visualize long-term deformation trends. These maps were plotted using `ps_plot('v-dso',1,[-30,30])`, where the wider colorbar better represents real displacement magnitudes observed in the area, including significant uplift and subsidence events.

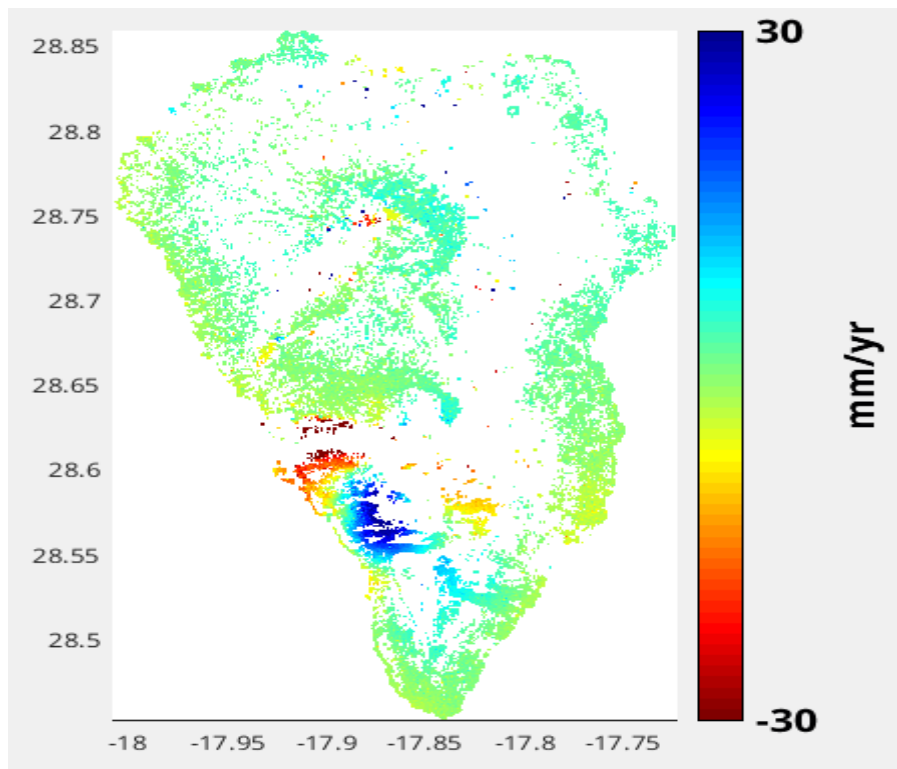


Fig17. Final corrected mean velocity map

The final mean velocity maps show consistent patterns of uplift and subsidence in targeted areas. Corrections applied in previous steps significantly enhanced the visibility of these trends.

Time Series Analysis (Step 17–18)

Following the refinement of velocity results, Steps 17 and 18 focused on analysing the temporal displacement behaviour at selected Persistent Scatterer (PS) points to validate long-term deformation trends and detect any anomalies. Using the command `ps_plot('v-dso',1,[-30,30],'ts')`, time series plots of Line-of-Sight (LOS) displacement were generated for carefully selected PS points exhibiting minimal phase noise. To ensure spatial consistency, a radius of 100 meters was used around each PS point, averaging the displacement behaviour of nearby scatterers for smoother and more representative trends.

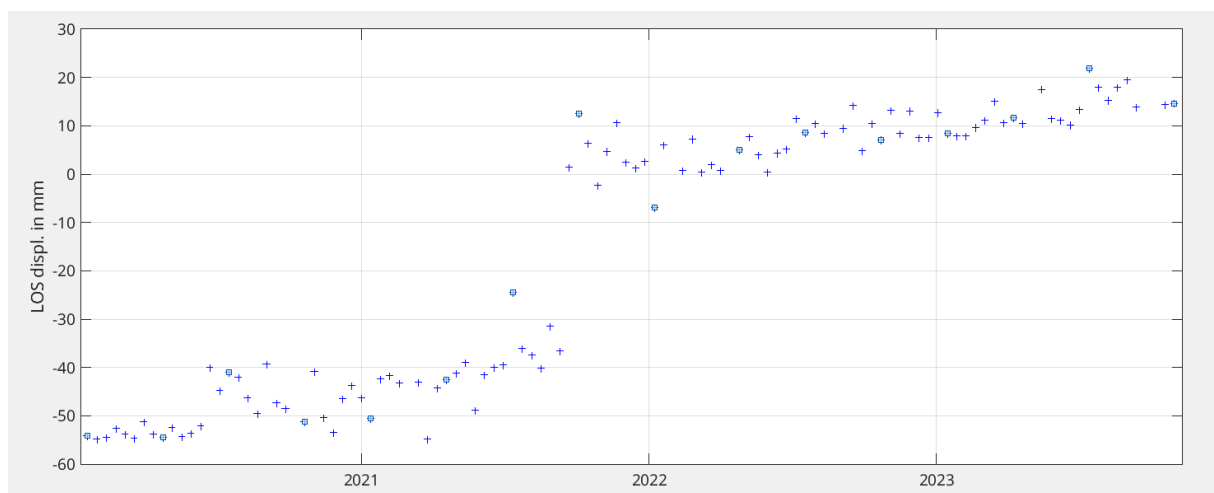


Fig18. LOS displacement time series

The time series shown corresponds to a point located in the southwestern uplift zone (marked in the velocity map), which experienced a sharp vertical displacement beginning in late 2021, correlating with the volcanic eruption at Cumbre Vieja.

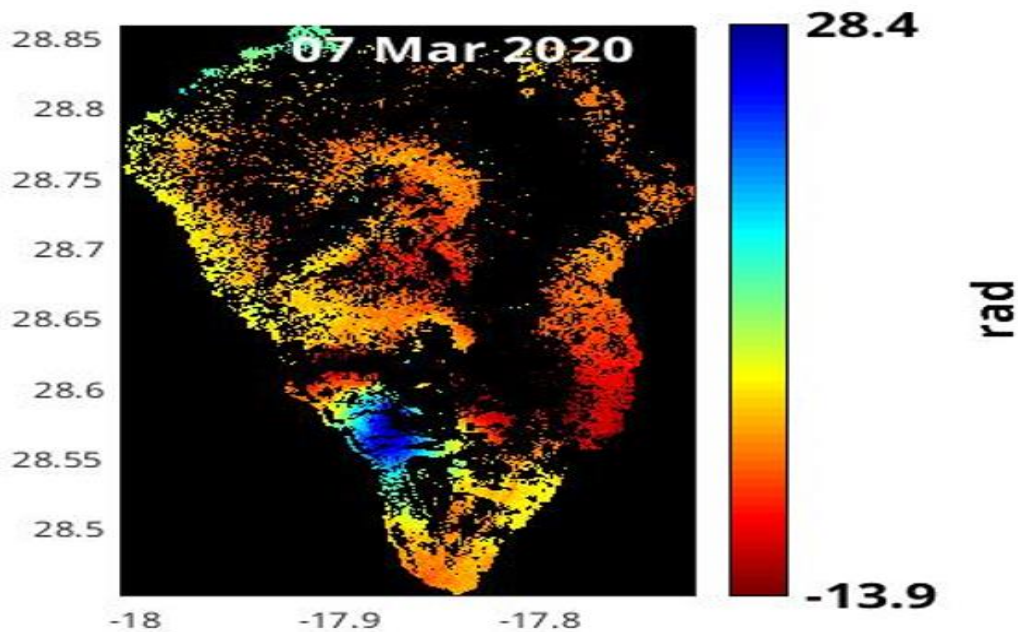


Fig19. Step displacement – event-related anomaly

Additionally, Fig. 19 showed a sudden displacement step, likely related to volcanic or seismic activity, which introduced a notable shift in ground movement during that period.

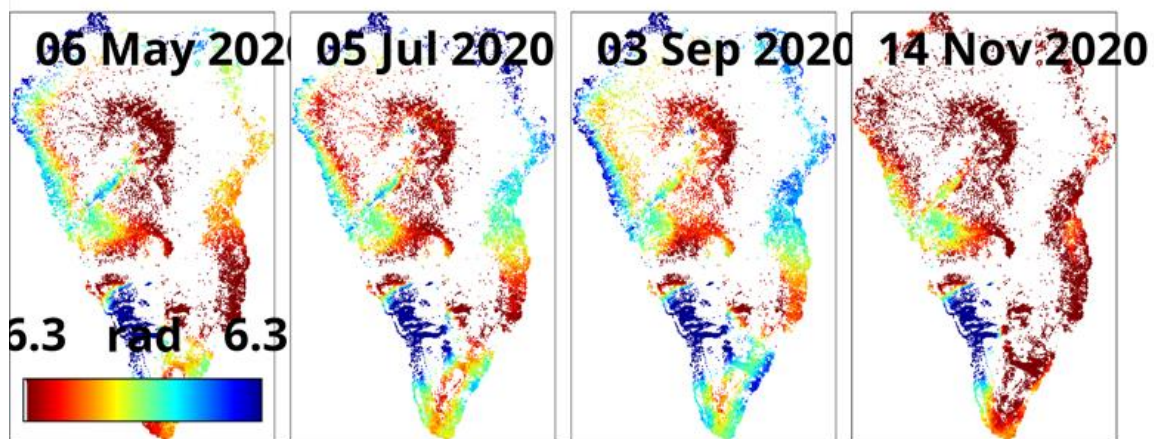


Fig20. Unwrapped phase – origin of displacement trends

To contextualize the source of displacement, the unwrapped phase map in Fig20 was included. This map helped trace observed displacements back to their phase origins, thereby linking the LOS displacement trends to real ground movement or processing artifacts.

Final Visualization and Results Compilation (Step 19)

In the final step, all corrected and validated outputs were compiled for comprehensive visualization and interpretation. The corrected mean velocity maps, shown in Fig. 15 (v-dso_2a) and Fig. 16 (v-dso_2b), provided a clear and accurate representation of long-term surface deformation across La Palma. These maps, corrected for atmospheric and orbital effects, revealed well-defined patterns of uplift and subsidence, consistent with known volcanic and tectonic activity in the region.

To further analyse cumulative LOS displacement over time, additional maps were generated, such as Fig. 13 (u-dmos_0.png) and Fig. 14 (u-dmos_2b.png). These maps visualized accumulated displacement and were critical in understanding deformation progression over the entire observation period.

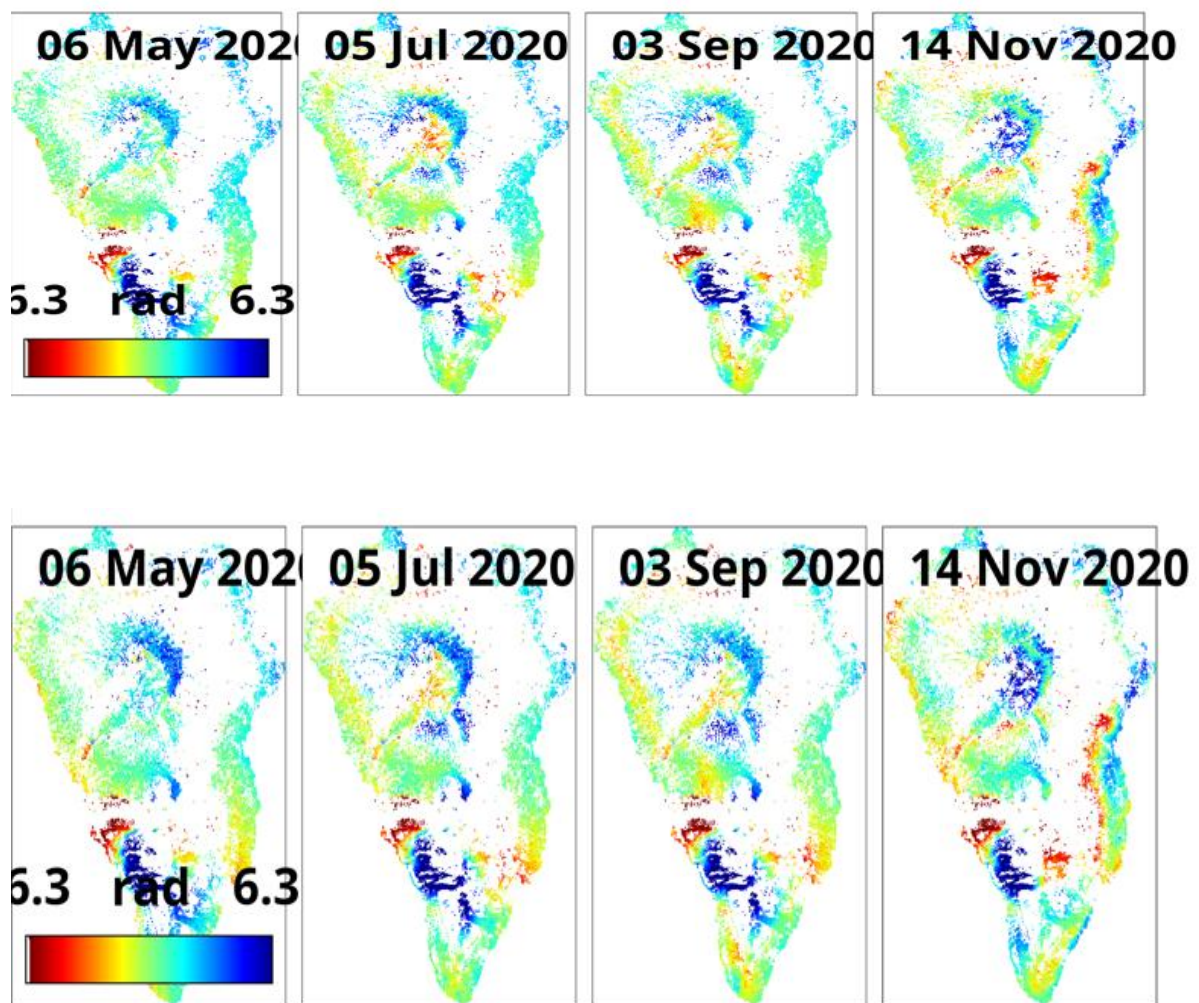


Fig21,22. LOS displacement map -cumulative deformation- alternative view

Together, these results provided a complete and validated picture of surface deformation. The combination of corrected velocity maps, detailed LOS displacement data, and thorough outlier assessment ensured high confidence in the PS-InSAR analysis outcomes, successfully achieving the goal of accurate ground movement monitoring using SAR data.

Deformation Summary and Time Series Interpretation

The corrected PS-InSAR results revealed clear surface deformation trends on La Palma, with localized zones of uplift and subsidence. The mean velocity maps highlight persistent uplift in the southwestern regions, particularly around the Cumbre Vieja volcanic ridge. This is consistent with known volcanic activity and subsurface magmatic processes in the area. Conversely, surrounding regions show moderate subsidence, possibly due to ground compaction or magma evacuation following the eruption. Uplift in key volcanic zones reaches up to 30 mm, while peripheral areas experience gradual displacement in both directions.

The time series plot further confirms these patterns. It shows a stable displacement trend at around -55 mm through most of 2021, followed by a distinct upward shift of approximately 30 mm during the late 2021 eruption, bringing the LOS displacement to about -25 mm. After this event, a slow and continuous uplift continues into 2023, reaching around +20 mm. This behaviour reflects typical volcanic inflation and post-eruption recovery. The selected point is located outside the peak uplift zone, allowing for clear observation of the transition from subsidence to uplift. These insights confirm the robustness of PS-InSAR for tracking both gradual and abrupt ground deformation, reinforcing its value in volcano monitoring and early warning systems.

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